

Ferdia Matti Treacy

📍 Dublin , Ireland ✉ ferdiatreacy@gmail.com ☎ 087 7842 759 in Ferdia Matti Treacy

🔗 My Github 📁 My Portfolio

Education

- BSc Maynooth University**, Computer Science and Software Engineering Sept 2023 – May 2027
- **President** of Maynooth Computer Science Society: 2026 - 2027
 - Year 3 : 94.0% (Ranked **First** in CS course)
 - Year 2 : 86.8% (Ranked **Second** in CS course)
 - Year 1 : 88.3% (Ranked **Third** in CS course)
 - (See Appendix for result breakdown)
- LC St Josephs Rochfortbridge** , Leaving Cert. Sept 2017 – May 2023

Employment

Intel NPU, AI SDE Intern. (*Pre silicon/Hardware Modelling team*)

- **Optimized hardware model accuracy from 72% to 100%** in testbench alignment. Closed critical gaps between the pre-silicon model and RTL implementation.
- Architected and deployed a **custom testing framework, now used throughout the team**, streamlining the identification of edge-case discrepancies.
- Collaborated closely with the pre-silicon validation, and RTL modelling team to debug complex architectural behaviors.
- Developed **CI pipelines** with github actions for automatic builds, deployment, and uploads to the share NFS of built C models.
- Implemented **test coverage and memory safety checking features**. Created custom CI pipelines with automatic artifact creation of specialized reports
- **Tools Used:** C, C++, Python, SystemVerilog, Github Actions, Synopsys VCS , Simics

Projects

Subtitle Translator - Microsoft Mentorshipped Project

[Subtitle Translator](#) 

- Collaborated in a team of 6 to build a full stack web application that takes in movie length subtitle files, parses and translates them with editing capabilities using an agile framework within an 8 week deadline. Mentored by Microsoft.
- Implemented Azure technologies such as AI voice recognition and Live Translation, used Azure for DevOps.
- Assisted in coding the backend, which utilized MongoDB and NPM packages such as express and subtitle.
- **Tools Used:** Javascript, Css, React, NodeJS , Azure Web Services, Jira, MongoDB.

Gameboy Emulator

[Gameboy Emulator](#) 

- Built a Game Boy emulator in modern C++ using only SDL3 and the C++ STL library.
- Passed all CPU testing; supports basic MBC and real-time rendering via SDL3.
- **Tools Used:** C++ , Assembly , SDL3 , CMake , Visual Studio Debugger.

EtymologySearch

EtymologySearch [↗](#)

- Created a React-based web application that allows users to input a sentence and highlights the etymology origin of each word.
- Built a Restful Flask API that performs web scraping on www.etymonline.com using BeautifulSoup4 to web scrape etymology data for words. Hosted on Heroku
- **Tools Used:** Python, Flask, Heroku , Javascript, Css, Html.

Game Jam Submission - "In the loop"

In the loop [↗](#)

- Created a game in Godot in 3 days for the GMTK 2025 Jam, the **biggest event ever** held on itch.io.
- Built the game from the ground up in **Python** and the Godot game engine.
- Placed **97th (top 1%)** in the narrative category, out of 9,593 entries and 37,000 participants
- **Tools Used:** Godot, GDscript(Python)

Volunteering

Maynooth University, Computer Science Center Tutor

Maynooth , Kildare Sept
2024 – Nov 2024

- Supported first-year students in mastering Java programming, focusing on foundational concepts and object oriented programming.
- Broke down complex problems into manageable steps, teaching students a structured approach to writing and organizing code to solve practical problems.
- Collaborated with other tutors to continually improve tutoring approaches, focusing on consistent and effective support for students learning needs.

Achivements

President of Maynooth Computer Science Society: 2026-2027

Maynooth University Hackathon December 2024 : 1st

Maynooth University Hackathon May 2023 : 5th

Appendix

Third year modules

Computation and Complexity	99%
Computer Networks	89%
Programming Languages and Compilers	96%
Software Design	100%
Software Verification	92%
Group Project	88%

Second year modules

Databases	93%
Algorithms & Data Structures 1	93%
Algorithms & Data Structures 2	97%
Computer Architecture	96%
Web Information Processing	97%
Operating Sys: Communications & Concurrency	91%
Software Testing	71%
Into User Experience (UX), User Interface (UI)	74%
Software Engineering & Software Process	91%
Theory of Computation	69%
Calculus 3 (S)	78%
Introduction to Statistics (S)	91%

First year modules

Introduction to Computer Science I	90%
Introduction to Computer Science II	100%
Computer Systems I	88%
Computer Systems II	92%
Mechanics & Special Relativity	90%
Mechanics & Modern Physics	71%
Differential Calculus	83%
Integral Calculus	96%
Linear Algebra 1 (S)	84%